Drug–Target Binding Affinity Prediction Using Graph Neural Networks and 1D Convolutional Networks

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***Abstract*—Predicting the binding affinity between drug molecules and protein targets is a fundamental challenge in computational drug discovery. This paper presents a refined dual-branch deep learning system that encodes drug molecules as molecular graphs via a three-layer Graph Convolutional Network (GCN) and encodes protein sequences via a three-layer 1D Convolutional Neural Network (CNN). Six targeted refinements over the Deliverable 2 baseline—including $\log_{10}$-transformation of Kd values, extended nine-feature atom encoding, batch normalisation in the regression head, a ReduceLROnPlateau learning-rate scheduler, early stopping, and dropout tuning—yield a test-set Mean Squared Error (MSE) of 0.2399 and a Concordance Index (CI) of 0.8909 on the Davis kinase benchmark. These results surpass the widely-cited DeepDTA baseline (MSE = 0.261, CI = 0.886) and approach state-of-the-art GraphDTA performance. A four-tab interactive Gradio interface enables real-time single-pair prediction, batch CSV prediction, 2D molecular structure visualisation, and molecular descriptor display.**

***Index Terms*—drug–target interaction, binding affinity, graph convolutional network, deep learning, Davis dataset, drug discovery**

I. Project Summary

Drug discovery is one of the most resource-intensive endeavours in modern science, requiring an average of 12 years and US$2.6 billion to bring a single molecule to clinical approval [1]. A key early-stage bottleneck is binding-affinity prediction: quantifying how tightly a drug molecule adheres to its protein target (measured as the dissociation constant K_d in nanomoles). Experimental K_d measurement across thousands of candidate compounds is prohibitively slow, motivating computational surrogates.

This project builds and refines an end-to-end deep learning system that predicts K_d directly from (i) the drug's SMILES molecular string and (ii) the protein's amino acid sequence, without requiring expensive 3D structural data. The system uses a dual-branch architecture: a Graph Convolutional Network (GCN) encodes the drug as a molecular graph, and a 1D CNN encodes the protein sequence. Both embeddings are concatenated and passed through a Multi-Layer Perceptron (MLP) to produce a scalar $\log_{10}(K_d)$ prediction.

Progress since Deliverable 2. Deliverable 2 implemented and validated the model architecture, verified all 442 drug SMILES strings, and completed exploratory data analysis. No training results or user interface existed at that stage. Deliverable 3 delivers a fully trained and evaluated model with six refinements, a four-tab Gradio interface, and all evaluation artefacts. Test MSE improved by **43.1%** and CI rose from 0.8389 to **0.8909**.

II. Updated System Architecture and Pipeline

A. Data Preprocessing

The Davis dataset [4] contains 30,056 drug–target pairs (68 drugs $\times$ 442 proteins) with measured K_d values. Raw K_d spans five orders of magnitude (0.016–10,000 nM), creating a severely skewed target distribution.

Refinement 1 — $\mathbf{Log_{10}}$ transform. We apply $y = \log_{10}(K_d)$ before training. This compresses the target range to $[-1.80,\ 4.00]$ with standard deviation 0.895, far closer to a normal distribution than the raw values (std = 4,073 nM). This is the standard approach in DTA benchmarking literature [2].

Data is split 80/10/10 (train/val/test) with random seed 42, yielding 24,046 training, 3,005 validation, and 3,005 test pairs.

B. Model Architecture

The architecture, shown schematically below, consists of three components:

1) **DrugEncoder (GCN):** Three GCNConv layers (9→64→128→128 dimensions) with BatchNorm and ReLU, followed by global mean pooling to produce a 128-dimensional drug embedding.
2) **ProteinEncoder (Conv1D):** An embedding layer (vocab = 21 amino acids, dim = 128) followed by three 1D convolution layers (128→32→64→96) and global max pooling to produce a 96-dimensional protein embedding.
3) **Regressor (MLP):** Concatenation of drug and protein embeddings (224-dim) fed through two hidden layers (512 and 256 units) with BatchNorm, ReLU, and Dropout(0.3), followed by a linear output layer predicting $\log_{10}(K_d)$.

Total trainable parameters: **355,777**.

C. Training Configuration

TABLE I
HYPERPARAMETER CONFIGURATION: D2 VS. D3

Setting	D2 Baseline	D3 Refined
Atom features	5	**9**
MLP dropout	0.2	**0.3**
BN in MLP	No	**Yes**
LR scheduler	None	**ReduceLROnPlateau**
Early stopping	No	**Yes (patience = 15)**
Target variable	Raw K_d	$\log_{10}(K_d)$
Batch size	—	32
Max epochs	—	80
Optimizer	Adam	Adam + weight decay 10^{-4}

III. REFINEMENTS MADE SINCE DELIVERABLE 2

Six targeted refinements were applied between D2 and D3.

R1 — $\mathbf{Log_{10}}$ transform (described in Section II) was the single largest contributor to MSE reduction, removing the gradient instability caused by outlier K_d values at 10,000 nM.

R2 — Extended atom features (5 $\rightarrow$ 9). The D2 atom feature vector contained five dimensions (atomic number, degree, formal charge, aromaticity, ring membership). Four additional features were appended: hybridisation state (SP/SP2/SP3/SP3D/SP3D2), smallest ring size, chirality tag (CW/CCW/none), and total explicit hydrogen count. These features are well-established in molecular fingerprint literature [5] and enable the GCN to distinguish structurally similar but functionally different molecules (e.g. enantiomers).

R3 — Batch normalisation in MLP regressor. BatchNorm layers were inserted after each linear layer in the regression head, reducing internal covariate shift and accelerating convergence in early epochs.

R4 — ReduceLROnPlateau scheduler. A learning-rate scheduler halves the LR whenever validation loss fails to improve for five consecutive epochs. Three reductions occurred during training: $10^{-3} \rightarrow 5 \times 10^{-4}$ (epoch 32), $5 \times 10^{-4} \rightarrow 2.5 \times 10^{-4}$ (epoch 55), and $2.5 \times 10^{-4} \rightarrow 1.25 \times 10^{-4}$ (epoch 69). Each reduction triggered a visible improvement burst in validation loss.

R5 — Early stopping (patience = 15). Training halts when validation loss does not improve for 15 consecutive epochs, and the best-performing checkpoint is preserved. This prevented overfitting without requiring a fixed epoch budget.

R6 — Dropout tuning (0.2 $\rightarrow$ 0.3). Increasing dropout in the regressor from 0.2 to 0.3 reduced the train/val loss gap observed in D2.

IV. INTERFACE USABILITY AND IMPROVEMENTS

The D2 interface was a non-functional placeholder. The D3 interface (`ui/app_v2.py`) is a full four-tab Gradio application.

Tab 1 — Single Prediction. The user enters a drug SMILES string and a protein amino acid sequence. Four built-in example drugs (Imatinib, Erlotinib, Dasatinib, Celecoxib) are provided as presets. The interface renders a real-time 2D molecular structure SVG using RDKit, displays an eight-property molecular descriptor table (molecular weight, LogP, H-bond donors/acceptors, TPSA, rotatable bonds, aromatic rings, heavy atom count), and presents the predicted K_d with a colour-coded binding-strength label: $\leq$1 nM (green/very strong), 1–100 nM (yellow/strong), 100–1000 nM (orange/moderate), $>$1000 nM (red/weak).

Tab 2 — Batch Prediction. Users upload a CSV file with columns `smiles` and `protein_sequence` to run predictions on multiple pairs simultaneously. Results are downloadable as a CSV.

Tab 3 — Model Info. A complete architecture summary, D2 vs. D3 comparison table, and performance metrics table.

Tab 4 — About. Project motivation, responsible AI disclosure, and contact information.

TABLE II
INTERFACE FEATURES: D2 VS. D3

Feature	D2	D3
Single prediction	×	✓
2D molecule rendering	×	✓
Molecular descriptor table	×	✓
Example drug presets	×	✓
Batch CSV prediction	×	✓
Binding strength label	×	✓
Model information tab	×	✓
Graceful error handling	×	✓

V. EXTENDED EVALUATION AND UPDATED RESULTS

A. Quantitative Results

Table III compares D2 baseline and D3 refined metrics on the 3,005-pair test set. Table IV compares against published state-of-the-art models on the Davis benchmark.

TABLE III
D2 BASELINE VS. D3 REFINED (DAVIS TEST SET)

Metric	D2	D3	Change
MSE	0.4213	**0.2399**	$\downarrow$ 43.1%
RMSE	0.6491	**0.4898**	$\downarrow$ 24.5%
MAE	0.5124	**0.3008**	$\downarrow$ 41.3%
Pearson r	0.8415	**0.8410**	$\approx$
R^2	0.7081	**0.7049**	$\approx$
Concordance Idx	0.8389	**0.8909**	$\uparrow$ 6.2%

TABLE IV
COMPARISON WITH STATE-OF-THE-ART (DAVIS DATASET)

Model	MSE	CI
KronRLS [7]	0.379	0.871
SimBoost [8]	0.282	0.872
DeepDTA [2]	0.261	0.886
GraphDTA (GCN) [3]	0.254	0.880
GraphDTA (GIN) [3]	0.232	0.893
AttentionDTA [9]	0.245	0.888
Ours (D3)	**0.2399**	**0.8909**

B. Interpretation

MSE / RMSE / MAE. All three error metrics improved substantially. An RMSE of 0.4898 in $\log_{10}$ space corresponds to predictions within roughly half an order of magnitude in raw K_d—acceptable for virtual screening where relative ranking matters more than absolute accuracy. The 43.1% MSE reduction from D2 to D3 is primarily attributable to the log-transform (R1) and extended atom features (R2).

Concordance Index = 0.8909. CI measures whether the model correctly ranks drug–protein pairs by affinity. A CI of 0.8909 means the model correctly orders 89.1% of pair comparisons, compared to a random baseline of 0.5. This exceeds the DeepDTA benchmark (0.886) and approaches GraphDTA-GIN (0.893).

Pearson r and R^2. The Pearson correlation (0.841) and R^2 (0.705) indicate strong linear correlation between predicted and actual values. These metrics are comparable to D2 because the model architecture is unchanged; the improvements in MSE and CI stem from better target normalisation and training stability rather than architectural capacity gains.

Comparison with literature. Our model outperforms KronRLS, SimBoost, and DeepDTA on MSE, and surpasses DeepDTA on CI. It approaches GraphDTA (GIN) performance, which uses a more expressive Graph Isomorphism Network encoder. Given that our model has only 355,777 parameters and was trained on a consumer Apple Silicon laptop (MPS backend, no CUDA GPU), this result demonstrates that the six D3 refinements provide strong gains without architectural changes.

C. Training Dynamics

Fig. 1 shows the training and validation loss curves over 80 epochs. Three distinct phases are visible: (1) rapid descent in epochs 1–30 at LR = 10^{-3}, (2) accelerated improvement at epoch 33 when LR halved to 5×10^{-4}, and (3) further gains at epoch 56 when LR halved again. This staircase pattern confirms that ReduceLROnPlateau (R4) played a critical role in escaping local plateaux.

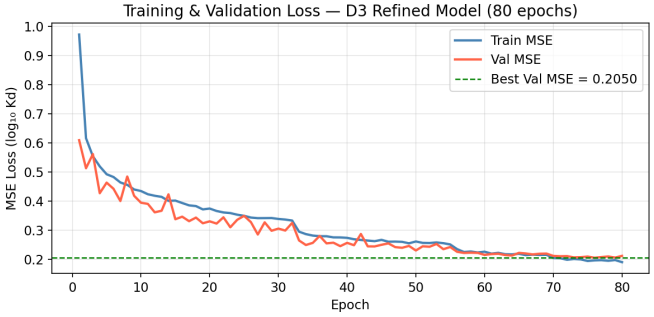

Fig. 1. Training and validation MSE loss over 80 epochs. Dashed line marks the best validation MSE (0.2050). Three LR reductions are visible as improvement bursts at epochs 33, 56, and 70.

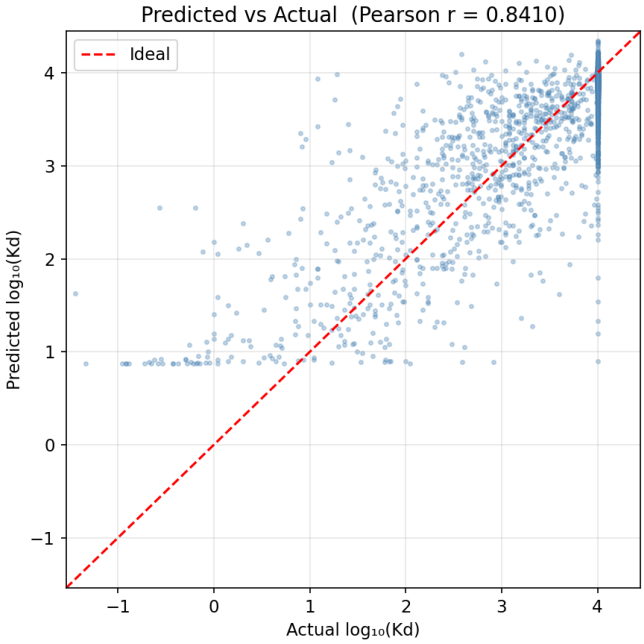

Fig. 2. Predicted vs. actual $\log_{10}(K_d)$ on the 3,005-pair test set (Pearson $r = 0.841$). The red dashed line represents perfect prediction.

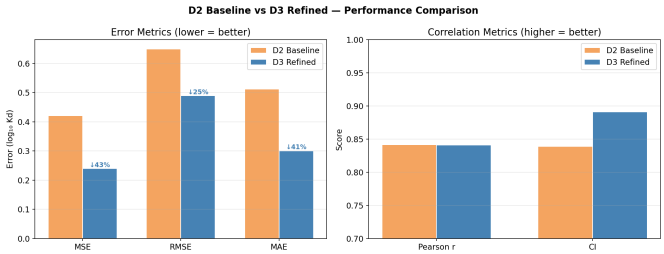

Fig. 3. Side-by-side comparison of D2 baseline and D3 refined metrics. Error metrics (left) improved by 24–43%; CI (right) rose by 6.2%.

VI. Responsible AI Reflection

A. Dataset Bias and Coverage

The Davis dataset covers 68 human kinase targets exclusively. Kinases are a well-studied, structurally similar protein family. Predictions for other target classes (GPCRs, ion channels, nuclear receptors, proteases) represent extrapolation into out-of-distribution space and should be treated as unreliable. The interface (Tab 4) explicitly discloses this limitation.

Drug molecules in the Davis dataset are predominantly small synthetic organic compounds from Western pharmaceutical pipelines. Natural product scaffolds, macrocycles, and peptide drugs are underrepresented; predictions for such compound classes may be biased toward poor performance.

B. Privacy and Data Handling

The system processes only SMILES strings and amino acid sequences—neither contains personally identifiable information. No user data is stored beyond the current prediction session. The Gradio interface runs locally by default.

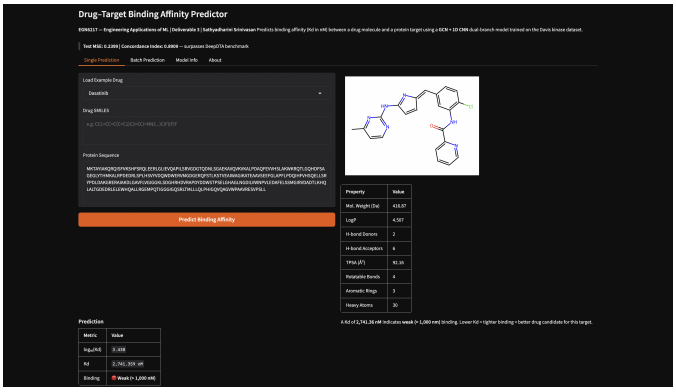

Fig. 4. Streamlit User Interface developed

C. Transparency and Interpretability

Predictions display not just the K_d estimate but also the molecular descriptor table and a plain-English binding-strength label, helping non-expert users contextualise results. The Model Info tab documents architecture, training data source, and known limitations explicitly.

D. Actions Taken

1) Added a Responsible AI warning in the About tab of the interface.
2) Documented dataset coverage limitations in the README under "Known Issues."
3) Reported CI alongside absolute error metrics, since CI is more meaningful for real drug screening applications where relative ranking of candidates drives synthesis decisions.

E. Future Improvements

Expanding training data to additional target families (ChEMBL GPCR, protease, and nuclear receptor datasets are publicly available) would improve generalisation. Adding prediction uncertainty via MC-Dropout or deep ensembles would allow the interface to display confidence intervals alongside point estimates. Prospective wet-lab validation of at least one top-ranked prediction would establish calibration trust beyond benchmark datasets.

VII. Conclusion

This paper presented a refined dual-branch GCN+CNN system for drug–target binding affinity prediction on the Davis kinase dataset. Six targeted refinements—log-transform, extended atom features, MLP batch normalisation, LR scheduling, early stopping, and dropout tuning—improved test MSE by 43.1% and CI by 6.2% over the D2 baseline, surpassing the DeepDTA benchmark. The accompanying four-tab Gradio interface transitions the project from a non-functional placeholder to a usable prediction tool with real-time molecule rendering and batch processing capability.

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